

Matematica - Clasa a 11-a

Exercitii - Toate capitolele

Nume: _____

Data: _____

Sectiunea 1 - Siruri si Progresii

1. Progresia aritmetica cu $a_1=3$ si $d=4$:

$$a_5=___ ; a_{10}=___ ; S_8=___$$

2. Progresia geometrica cu $a_1=2$ si $q=3$:

$$a_4=___ ; S_5=___ ; S_{-\infty} (q=1/2, a_1=4) = ___$$

3. Calculeaza limita:

$$\lim_{n \rightarrow \infty} (3n^2+2n)/(n^2-1)=___ ; \lim_{n \rightarrow \infty} n/(n+1)=___$$

Sectiunea 2 - Derivate si aplicatii

4. Calculeaza derivata:

$$(x^4-2x^2+1)'=___$$

$$(x \times \ln x)'=___ \text{ (produs)}$$

$$(\sin(x^2))'=___ \text{ (compusa)}$$

$$(e^{x/x^2})'=___ \text{ (cat)}$$

5. Studiaza $f(x) = x^3 - 6x^2 + 9x$:

$$f'(x)=___ ; \text{ puncte critice: } x=___ \quad x=___$$

Maxim local in ___ ; Minim local in ___

Crescatoare pe ___ ; Descrescatoare pe ___

Sectiunea 3 - Primitive si Integrale

6. Calculeaza primitivele:

$$\int (3x^2-2x+5)dx=___$$

$$\int \sin(2x)dx=___ \text{ (substitutie)}$$

$$\int x e^x dx=___ \text{ (parti: } u=x, dv=e^x dx)$$

7. Integrala definita:

$$\int_{[0,2]} x^2 dx=___ ; \int_{[0,\pi]} \sin x dx=___$$

8. Aria din grafic:

$$\text{Aria intre } y=x^2 \text{ si } y=x \text{ pe } [0,1]: A=\int_{[0,1]} (x-x^2)dx=___$$

Exercitii Clasa 11 (continuare)

Sectionea 4 - Matrice si Sisteme

9. Calculeaza determinantii:

$$\det[[2,3],[1,4]]=______ ; \det[[1,0,2],[0,1,3],[2,1,0]]=______$$

10. Inversa matricei $A=[[1,2],[3,4]]$:

$$\det(A)=______ ; A^{-1}=______$$

11. Rezolva prin Cramer:

$$3x+2y=7 \text{ si } x-y=1 \Rightarrow x=______, y=______$$

Sectionea 5 - Combinatorica si Prob.

12. Calculeaza:

$$C(8,3)=______ ; A(6,2)=______ ; C(10,4)=______$$

13. Termenul T_4 din $(2x-1)^6$:

$$T_4=C(6,3)(2x)^3(-1)^3=______$$

14. $X \sim B(5, 0,4)$. Calculeaza $E(X)$, $D(X)$, $P(X=2)$:

$$E(X)=______ ; D(X)=______ ; P(X=2)=______$$

Sectionea 6 - Adevarat / Fals

- $\int e^x dx = e^x + C$ ---
- $\det(AB) = \det(A) + \det(B)$ ---
- $f''(x_0) > 0 \Rightarrow \text{minim local}$ ---
- $(fg)' = f'g'$ ---
- $S_{\text{inf}} = a/(1-q)$ pt $|q| < 1$ ---
- $C(n,k) = C(n,n-k)$ ---

Exercitii Clasa 11 - Probleme

Sețiunea 7 - Probleme aplicate

15. Studiaza complet $f(x)=x^2 e^x$ (varf, monotonie, asimptote).

$$f'=xe^x(x+2) \quad f'=0: x=0, x=-2 \quad \text{Min in } x=0, \text{ Max in } x=-2$$

16. Calculeaza aria suprafetei dintre $f(x)=x^2$ si $g(x)=2x$.

$$\text{Intersectii: } x^2=2x \Rightarrow x=0, x=2 \quad A=\int_0^2 (2x-x^2) dx=---$$

17. O minge e aruncata sus: $h(t)=-5t^2+20t$. Inaltimea maxima?

$$h'(t)=-10t+20=0 \Rightarrow t=2 \quad h(2)=--- \quad \text{Aceasta e inaltimea maxima}$$

18. Intr-o urna: 5 rosii, 3 albastre. Extragi 2 fara inlocuire.

$$P(\text{ambele rosii})=C(5,2)/C(8,2)=10/28=---$$

19. Rezolva sistemul 3×3 : $x+y+z=6$, $2x-y+z=3$, $x+2y-z=2$

$$\text{Gauss: } [A/b] \rightarrow \text{forma treapta} \rightarrow x=---, y=---, z=---$$

20. Calculeaza $\int_0^1 x\sqrt{1-x^2} dx$ (substitutia $t=1-x^2$)

$$t=1-x^2 \Rightarrow dt=-2x dx \Rightarrow -1/2 \int_1^0 \sqrt{t} dt = ---$$

Exercitii Clasa 11 - Calcule rapide

Sectiunea 8 - Derivate rapide

$(x^5)' = ______$

$(e^{(3x)})' = ______$

$(\ln(2x))' = ______$

$(\cos(x^2))' = ______$

$(x^2 \sin x)' = ______$

$(1/(x+1))' = ______$

$(\sqrt{x^2+1})' = ______$

$(\tan x)' = ______$

Sectiunea 9 - Integrale rapide

$\int 5x^4 dx = ______$

$\int \cos x dx = ______$

$\int 1/x dx = ______$

$\int e^{(2x)} dx = ______$

Sectiunea 10 - Matrice rapide

21. Calculeaza:

$\det[[3,1],[2,4]] = ______ ; \det[[-1,2],[3,5]] = ______$

22. Produsul $A \times B$:

$A = [[1,2],[0,1]], B = [[2,0],[1,3]]: AB = ______$

Sectiunea 11 - Demonstratii

23. Demonstreaza: daca $a_n = 1/n$, atunci $\lim a_n = 0$.

24. Demonstreaza: $\int[a,b](f+g) = \int[a,b]f + \int[a,b]g$

Exercitii Clasa 11 - Test final

25. Progresia geometrica: $a_3=12$, $a_5=48$. Gaseste a_1 si q .

$$a_5/a_3=q^2=4 \Rightarrow q=2 ; a_3=a_1 q^2=4a_1=12 \Rightarrow a_1=3$$

26. Studiul semnului lui $f'(x)$ pt $f(x)=x^3-3x^2$.

$$f'=3x^2-6x=3x(x-2)=0 \text{ pt } x=0,2 \text{ Min in } 2, \text{ Max in } 0$$

27. Ecuatia tangentei la $f(x)=\ln x$ in $x=1$.

$$f(1)=0; f'(x)=1/x; f'(1)=1 \text{ Tangenta: } y=x-1$$

28. $\int [1,e] (1/x + x) dx = \text{---}$

$$=[\ln x + x^2/2]_{-1}^e = (1+e^2/2)-(0+1/2) = 1/2+e^2/2$$

29. Probabilitatea Bayes: $P(A)=0.3$, $P(B/A)=0.8$, $P(B/A')=0.2$.

$$P(A/B) = P(B/A)P(A) / [P(B/A)P(A)+P(B/A')P(A')] = \text{---}$$

30. $X \sim B(10, 0.3)$. $P(X \geq 1) = ?$

$$P(X \geq 1) = 1 - P(X=0) = 1 - C(10,0)(0.3)^0(0.7)^{10} = 1 - 0.7^{10} \approx \text{---}$$